



Assignment 5

1 Instructions

- **Deadline:** 24 hours before next laboratory class.
- **Before implementing**, draw **UML Class Diagrams**, save them as images and **copy each one into the root of their projects**.
- Rename projects. Export them to a **single** archive file `surname1_surname2_name_lab05`.
- Rename package names `lab04` to `lab05`.

Whatever the exercise, be sure you always meet all the following rules:

1. **Write tests first.** When **updating** a project, check that existing tests keep working properly. Even more, include new characteristics in existing tests.
2. Review the **checklist** to check the correctness of your projects.

2 Adding Books to DOME

Take as starting point project DOME implemented in the previous laboratory session 4.

Add a new Book item. A book has

- a **title**, a **writer** that is also the **responsible**, a **publisher** and an **ISBN**, such as A49290
- a **base price**.
- As other items, books may be **owned** or not and a **comment** can be added to them.
- The base price will be increased by 4% to calculate the **final price**.
- Books are equal if they have the same title and ISBN.

For all them, null, empty and blank strings are not allowed, and only numbers greater than zero are accepted.

From now on, books and CD (**but neither DVD nor video games**) may be **borrowed** and **given back**, and availability can be checked. They will be available whenever they are **owned and not borrowed**, assuming there is only one copy of each.

There will be three methods to implement these new characteristics just for CD and Book instances.

- `void borrow()`
- `void giveBack()`
- `boolean isAvailable()`

Mind the difference between borrowable (CD and Book) and available (CD and Book but only when they are owned and not borrowed).



2.1 New methods in class MediaLibrary

First of all, write tests for methods `borrow`, `giveBack` and `getBorrowables`, described below.

- **getBorrowables:** Returns a String with all borrowable items (cds and books). Each item will be separated from next with newline character `'\n'`.
- **getAvailables:** Returns a String with all available items. Each item will be separated from next with newline character `'\n'`.
- **borrow:** It borrows a CD or Book (not null) received as argument. Then, it looks for an item in the list equals the argument and marks it as borrowed, if possible. Returns the borrowed item or null, if it was impossible to borrow the item.
- **giveBack:** Receives an argument, Book or CD, not null. Then, it looks for an item in the list equals the argument and marks it as available, if possible. Returns true if it was possible to give the item back. False, otherwise.

2.2 Update MediaPlayer

Finally, updates `MediaPlayer` by adding five books (2 owned), 3 CD (2 owned).

Then, borrow one book and one cd.

Print all borrowables and all availables.

3 Shapes

Using the project shapes as starting point,

- Add a class `Circle` with center, radius and colour
- Add a class `Triangle` (right-angled) with x and y, leg, angle and colour

Then, in project `picture`, add a new class `Picture` with x, y, height, width and filename, that can be drawn in the console with the following message:

```
Drawing Picture filename (x, y):  heigh:  height, width:  width
```

Complete the tasks that have not been done in the lab.